Langfuse: What Stays Borrowed When Strangers Arrive?

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MODULE 06 · PROTOTYPING WITH AI · CASE, READ BEFORE CLASS

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On 29 August 2023, at 16:14 UTC, Marc Klingen posted “Show HN: Langfuse – Open-source observability and analytics for LLM apps” to Hacker News, a forum where software developers discuss and vote on new projects.[1] The post linked the code on GitHub, a two-minute video and a live demo. Langfuse recorded what happens inside applications built on large language models, each request and its steps as a trace, so that developers could see and improve them.[1] Behind the post stood 15 weeks of work by Klingen, Max Deichmann and Clemens Rawert, the three founders of Langfuse, and 300 commits, saved changes to the code, since the repository’s first day, 18 May 2023.[2] [3]

Not everything a stranger from Hacker News would find was built by the founders. Every user saw the app’s own pre-built dashboards.[1] [4] The analytics page, which the post called the “analytics MVP”, ran on a report in Looker Studio, Google’s tool for building reports, embedded in the app and shown only to users whose account carried a flag in the database.[1] [4] Every other user saw a notice that analytics were “in closed alpha”.[4] In the post itself, Klingen named the database and Looker Studio as the “two engineering decisions we’ve felt uncertain about”.[1] The question in front of the founders was which of the parts they had borrowed or kept behind a flag to replace first, and what to build next.

Logging Prompts to the Production Database

According to the company’s handbook, the three founders had started working together in 2022 on Finto, a product for usage-based billing, with which they joined the Winter 2023 batch of Y Combinator, the startup accelerator in San Francisco, from January to April 2023.[5] Klingen recalled in 2025 that they decided in the first weeks of the batch to drop it.[6]

During the batch the founders built applications on large language models. “When GPT-4 dropped, we started building LLM apps – a lot of them!”, Klingen wrote on 29 August 2023, and linked one of them, learn-from-ai.com.[1] By the founders’ later accounts, they tried several such ideas, most for less than two weeks each, among them AI-generated courses and a bot that turned GitHub issues, the bug reports and requests on a code repository, into pull requests, proposed code changes.[5] [7] The founders told an interviewer in 2024, “We realized our demos were cool, but back then it was hard to make this useful in practice.”[7]

Every one of these applications, Klingen wrote, “suffered from the same issue: it’s hard to assure quality in 100% of cases and even to have a clear view of user behavior.”[1] “Initially, we logged all prompts/completions to our production database to understand what works and what doesn’t,” he continued. “We soon realized we needed more context, more data and better analytics to sustainably

improve our apps. So we started building a homegrown tool."[1] In November 2023 the founders added that they had first worked on a test framework for the outputs of language models and had "quickly realized that tests give a false sense of security".[8] When that happened is not documented.

The batch's habit, as the founders recalled it in 2024, was to ask "why not try building it in two days instead of a week?"[7]

Fifteen Weeks of Building

Max Deichmann created the repository on 18 May 2023 at 17:47 UTC, after the batch.[9] [5] Its first commit held a template for several apps, whose README carried another name. A commit called "kill all" deleted 51 of its files, and an "init" commit 81 minutes after the first one started a single web app named "langfuse".[10] Why the founders started over that evening is not documented. Within 24 hours the log showed a database schema, a "trace api", "observations", a frontend and "sdks" (Exhibit 1).[2]

Klingen recalled in 2025 that the first prototype was a hack of about a week. The founders sent it to a few companies they knew from the batch as a page in Notion, the note-taking tool, used as a website, with a link to the code on GitHub to try it.[6] Between two and five users tried it, by his account, and the founders spent two more weeks asking them why they used it, since, as he put it, if it took the founders a week, it might take the customers two.[6] In their November 2023 account the founders wrote that they "tested it with other founders", and "After very positive initial feedback, we decided to open-source the project".[8] The repository was public by 10 June 2023 at the latest.[11]

A commit named "v1" landed on 26 May, day 9. That day's README stated the goal as "Debug and improve your LLM-based application by logging/analyzing all user interactions, backend traces, and explicit/implicit user feedback", offered a local start with Docker, a tool that packages software to run on any machine, and put the code under the MIT licence, except one folder.[12] Accounts, projects and a hosted cloud version followed on 1 June.[13] A README of 8 June marked four features as "coming soon".[14]

The founders described their tools on 29 August. "Under the hood, we use the T3 stack (Typescript, NextJs, Prisma, tRPC, Tailwind, NextAuth), which allows us to move fast + it means it's easy to contribute to our repo," Klingen wrote. The software development kits (SDKs), the small libraries developers add to their own code to send data to Langfuse, were "heavily influenced by the design of the PostHog SDKs", an analytics company's open-source libraries. For converting the API description into Python code the founders used Fern, a code generator, and for tables and dashboards ready-made components.[1] The cloud version ran its database on Supabase, a hosted Postgres service, and the app on Vercel, a hosting platform.[1] The marketing page ran on Framer, a website builder, until 20 August.[15]

Klingen and Deichmann wrote almost all of the code. Of the 300 commits before the post, Klingen authored 159, Deichmann 130, Rawert 9 and outside contributors 2, and commits fell on 68 of the 104 days.[2] The website and documentation lived in a second repository with 317 commits by the morning of 29 August (Exhibit 2).[16] A build pipeline was added on 28 May. The first automated tests came on 10 August and ran in that pipeline from 17 August, day 92.[2] [17]

From a Call with a Founder to a Stranger on Their Own

On 7 June the "No account yet?" link in the app led to an external sign-up form, and the "Sign up for early access" button on the website led to a page where visitors booked a call with a founder.[18] On 26 June the founders opened sign-up to everyone. The app's start screen then offered two cards. One

read “Get started by yourself. You can always reach out to us later or get help in the discord community.” The other, “Guided onboarding”, read “If you prefer 1:1 support, we are happy to help” and offered a call with a founder.[19] Discord, named on the first card, was a chat platform on which the founders ran a community server.[12] Both cards, and a third that opened the demo project, stood on the start screen on 27 August.[20] Klingen recalled in 2025 that the batch had ended with only a prototype and that about two months later “we felt like, okay, now, like a team that doesn’t know us can pick this up.”[21]

Around the product the founders added ways to see what users did. From 23 June a feedback button in the app sent comments to the founders’ error tracker and to their Slack.[22] From 18 July PostHog, a product analytics tool, recorded use of the web interface.[23] From 29 July cloud users saw a support chat, because, as Klingen wrote in the change, “we want to offer them support proactively while setting up their project.”[24] From 14 August every new user got view-only access to a demo project with live data.[25] The data in that project came from a chatbot that answered questions about the documentation. Klingen had built it in July after “A user was surprised when I (a human) answered his/her question. It’s 2023, a bot was expected.”[26]

An open issue of 21 August stated that “Users cannot change or reset the password.”[27] Exhibit 3 lists what a new user met on 29 August.

What Users Did With It

The founders told an interviewer in 2024 that “One big unlock of YC for us was actually finding that first set of users”.[7] The company’s undated handbook describes the first customers as teams from the batch who spent much time in the product and referred others.[5] On 29 August the landing page listed five teams under “Teams building complex LLM apps rely on Langfuse”, Fastgen, Langdock, Alphawatch, Berry and Nucleus.[28]

The founders reported what users asked for. Their July update said that most users integrated through the Python or JavaScript SDKs and that many wanted to score outputs by hand in the interface, which shipped on 31 July.[29] In a reply in the Hacker News thread after the post on 29 August, about the use they had seen, Deichmann wrote that one user had described the difference to other tools this way, that “they use observability usually to know that nothing is going wrong whereas they use Langfuse many hours per day to understand how to best improve the application”.[30] No figure for sign-ups, active users or returning users before 29 August was published.

What counted as a launch varied. Sign-up opened on 26 June, the founders’ post on Y Combinator’s launch page described Langfuse on 19 July as “open-source product analytics (think ‘Mixpanel’) for LLM apps”, and on 20 August Langfuse was the day’s top product on Product Hunt, a site where makers present new products.[31] [32] Fifteen minutes after the Hacker News post, the GitHub page showed 370 stars, the bookmarks GitHub users give a repository, and 21 copies made by others.[3]

Borrowed and Behind a Flag

The analytics page had embedded the Looker Studio report since 17 July. Users with the flag “analytics-alpha” saw it. The flag sat in a column of the users table, and the change that added it added no screen in the app to set it.[4] [33] “Currently we are working with a small group of alpha users to test our analytics features,” the July update said, and “We are onboarding new users every week.”[29] The banner for alpha users read “If you cannot access the Looker-powered dashboards, please sign into your Google Account in another tab and try again. A version for smaller screens is not

yet available."[4] Every user, flagged or not, saw the app's own dashboards with traces, token usage, scores and per-user views, which Klingen showed in the two-minute video.[34] [4] [25] Token usage was the amount of text a model processed, counted in tokens.

The documentation explained the alpha. "We rely on some closed source software to make this work in order to move fast and focus on what's most interesting for most of you. We'll soon bring these features to all users since we currently plan a migration to an OLAP database and flexible modeling structure to power these features."[35] An OLAP database is one built for fast analytical queries over large amounts of data.

In the post Klingen wrote, "Integrating Looker Studio got us to first analytics charts in half a day. As it is not open-source and does not work with our UI/UX, we are looking to switch it out for an OSS solution to flexibly generate charts and dashboards." He named Lightdash, an open-source tool, as one they had looked at.[1] About the database he wrote that "Supabase performs well at our scale and integrates seamlessly into our tech stack."[1]

The SDKs were the founders' own. The post did not mention OpenTelemetry, an open standard for tracing software. In a reply later that day, about the state at the time, Deichmann wrote that they had "aligned our domain language to OpenTelemetry" and that "Currently users add lots of metadata and configuration details to the trace by manually instrumenting it using the SDKs (or via Langchain integration)." He added, "We are thinking about integrating OpenTelemetry, as this would be a step function on making integrations with apps easier. However, hadn't had the time yet to figure out how capture all the metadata that's relevant as context to the trace."[36] In another reply he explained the choice of their own SDKs. "We wanted to have a product out there fast and liked the experience of e.g. Posthog SDKs."[37]

16:14 UTC

The post ended with the founders' plan. "Next, we're going deep on analytics. For quality specifically, we will build out model-based evaluations and labeling to be able to cluster traces by scores and use cases."[1]

Two engineers had built a working product in 15 weeks, partly from parts they had not built. A borrowed report carried the analytics behind a flag that no screen in the app could set. A call with a founder still sat next to self-sign-up. The SDKs were their own, built fast. Strangers were about to arrive. Which of these parts should the founders replace first, and what should they build next?

Exhibit 1: Langfuse from First Commit to Show HN, 18 May to 29 August 2023

Date (2023)	Day	Event
18 May	1	Repository created, first commit, “kill all”, a fresh “init”, database schema
19 May	2	“trace api”, “observations”, frontend, “sdks”
25 May	8	Website and documentation repository, first commits and “publish”
26 May	9	Commit “v1”, README states the goal, local start with Docker, MIT licence
28 May	11	Build pipeline added
1 June	15	Accounts, projects and a hosted cloud version
7 June	21	Sign-up through an external form, website button books a call with a founder
By 10 June	24	Repository public (oldest surviving outside copy)
23 June	37	Feedback button and error tracking
26 June	40	Public sign-up, start screen with “Get started by yourself” and “Guided onboarding”
17 July	61	Analytics page with the Looker Studio report for flagged users
18 July	62	PostHog records use of the web interface
19 July	63	Y Combinator launch page
29 July	73	Support chat for cloud users
31 July	75	Manual scoring in the interface, documentation chatbot
10 August	85	First automated tests
14 August	89	Demo project for every new user
17 August	92	Tests run in the build pipeline
20 August	95	Landing page moved off Framer, top product of the day on Product Hunt
29 August	104	Show HN at 16:14 UTC

Source: compiled by the case writer from the commit logs and changes of both repositories, the founders’ posts and the Product Hunt record, see notes [2], [16], [10], [12], [13], [18], [11], [22], [19], [4], [23], [31], [24], [29], [26], [17], [25], [15], [32] and [1].

Exhibit 2: Commits per Week, 18 May to 29 August 2023

Week (2023)	App repository	Website and documentation repository
15 to 21 May (from 18 May)	34	0
22 to 28 May	63	5
29 May to 4 June	25	45
5 to 11 June	6	15
12 to 18 June	5	11
19 to 25 June	28	17
26 June to 2 July	20	12
3 to 9 July	11	4
10 to 16 July	13	12
17 to 23 July	19	17
24 to 30 July	12	34
31 July to 6 August	16	34
7 to 13 August	11	27
14 to 20 August	19	50
21 to 27 August	18	27
28 to 29 August (to 16:14 UTC)	0	7
Total	**300**	**317**

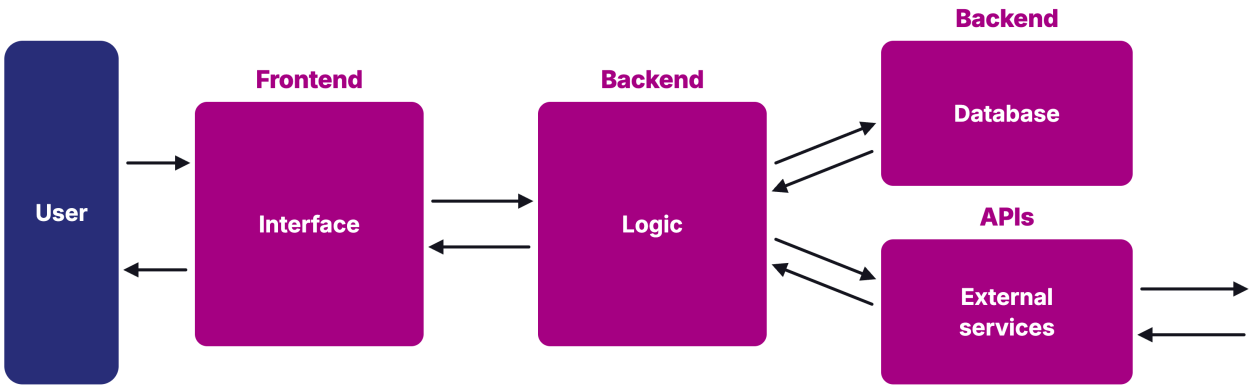
Source: case writer's count from the commit logs of the main branches of langfuse/langfuse and langfuse/langfuse-docs, author dates in UTC, weeks Monday to Sunday. Merged changes count once, so the numbers understate the work. See notes [2] and [16].

Exhibit 3: What a New User Met on 29 August 2023

What	State on 29 August 2023	Since
Website	Landing page in the documentation site, five named teams	20 August (off Framer)
Sign-up	Public, with email and password	26 June
Start screen	“Create new project” (get started by yourself), “View demo project” and “Guided onboarding” (a call with a founder)	26 June, as on 27 August
Demo project	View-only, live data from the documentation chatbot	14 August
Documentation	Quickstarts for cloud, local and self-hosted use, Python and JavaScript SDKs, a Langchain integration (“full integration in 5 minutes”, Langchain being a code library for building such applications), a public API	By 29 August
Traces	A viewer for each request and its nested steps, with user feedback	Since the first weeks, nested view 24 July
Dashboards	Traces, token usage by model, scores, per-user views, for every user	By 27 August (video)
Manual scoring	Scores set by hand in the interface	31 July
Analytics (Looker Studio)	For users with the flag “analytics-alpha” only. Others saw “Analytics is in closed alpha” and an invitation to get in touch	17 July
Support	Feedback button, support chat offered while setting up a project, a Discord community	23 June, 29 July
Account	No way to change or reset the password	Open issue of 21 August

Source: compiled by the case writer from the app’s code changes, the documentation site as committed on the morning of 29 August 2023 and the two-minute video, see notes [15], [28], [19], [20], [25], [38], [34], [29], [4], [22], [24] and [27].

Every Application Ever



Every application ever

Endnotes

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